# Mountain pine beetle: an example of a climate-driven eruptive insect impacting conifer forest ecosystems

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## Abstract

Climate change is altering the survival and reproductive capacity of plant-feeding insects in multiple ecosystems worldwide, in some cases creating conditions highly suitable for population eruptions. Forest ecosystems are particularly sensitive to climate change as their vulnerability is manifested, in part, as an upsurge in natural disturbances such as native insect outbreaks. The mountain pine beetle (MPB), *Dendroctonus ponderosae* Hopkins (Coleoptera: Curculionidae), is a phloem-feeding bark beetle indigenous to western North America that attacks most species of pine including its major hosts, lodgepole pine and ponderosa pine. Adult mass aggregation, mediated by pheromones, helps the beetle to overcome tree defenses eventually killing the tree. Recent outbreaks of this insect have caused extensive pine mortality and have affected millions of hectares of forested area in western North America. Climate is a major driver of these outbreaks. In this review, we describe the direct influences of various climate-related factors on MPB development, outbreak behavior, and range expansion and their indirect impact on MPB epidemiology via influences on host trees and MPB-associated fungi. We also underscore the ecological and economic consequences of the recent, unprecedented MPB outbreak. Of serious concern currently is whether climate change will facilitate rapid establishment and spread of MPB in naïve pine forests. MPB will likely adapt quickly to new thermal environments under climate change given its short generation time; however, uncertainties and gaps in our understanding of MPB population dynamics (e.g., trophic interactions) in newly invaded habitats preclude an accurate assessment of outbreak potential and spread at this time.

**Keywords:** climate, weather, mountain pine beetle, outbreak, range expansion, boreal, jack pine

**Review Methodology:** We used Elsevier's Scopus online database (https://www.scopus.com/search/form.uri?display=basic) to make a query using the search strings "mountain pine beetle" and "*Dendroctonus ponderosae*" with the Boolean operator 'OR' between the two search queries under "Article title, Abstract, Keywords" field on 13 July 2020. We searched "ALL" document types (articles, reviews, books, book chapters, etc.) irrespective of the access type and downloaded 1181 Scopus-indexed citations and abstracts, including manuscripts *in press*, from 2000 to July 2020. The initial list of 1181 citations was thinned by screening the title/abstracts to exclude duplicates and certain gray literature such as unrelated government documents, reports, theses, and conference abstracts. We considered some very pertinent review papers, symposium/technical reports, books, or book chapters as well as relevant publications within the articles in the screened list for citations. Additional web searches using multiple keywords (e.g., in Google Scholar) were made, as needed. A total of 167 publications were considered for inclusion in this manuscript.

## Introduction

Native bark beetles are important disturbance agents in the conifer forests of North America and Europe. They play key ecological roles in the functioning of a forest ecosystem both directly by maintaining a diverse forest landscape, that is, comprising different ages and compositions of tree stands, by helping remove diseased/stressed trees, and indirectly via their influence on ecosystem processes such as nutrient cycling, forest succession, fire dynamics, and hydrology. While most bark beetles do not cause serious damage to forests, certain "aggressive" bark beetles exhibit landscape-scale outbreak behavior that could decimate forests and lead to, in certain cases, biome-scale ecosystem changes and enormous economic impacts [1, 2]. The mountain pine beetle (MPB), *Dendroctonus ponderosae* Hopkins (Coleoptera: Curculionidae: Scolytinae), is one such native disturbance agent that is a natural part of the western North American conifer forests and whose outbreaks cause extensive mortality of pine trees including its main hosts, lodgepole pine (*Pinus contorta* Dougl. ex Loud. var. *latifolia* Engelm.) and ponderosa pine (*P. ponderosa* Dougl. ex Laws.). The aggressive nature of MPB attack refers to the pheromone-mediated mass attacks that help the beetles to quickly overwhelm the tree's defensive mechanisms and cause tree death within a year [1, 3] (Fig. 1A–C). Under high beetle pressure within a stand, fine-scale population eruptions occur, which eventually may spread over large areas (Fig. 1D). Although such localized or subregional outbreaks of MPB are common within its historical range, recent warming conditions and extensive availability of susceptible hosts on the landscape have helped the MPB to outbreak in the 2000s on a scale unprecedented in recorded history in North America and resulted in widespread range extensions and host shifts [4–7]. Expansive tree mortality caused by this outbreak has turned the forested region in British Columbia from a carbon sink to a net carbon source, potentially aggravating the effects of climate change [8].

In western Canada, outbreaking MPB populations originating from north-central British Columbia overflew the geoclimatic barrier of the Rocky Mountains and spread further northeastward into the province of Alberta, Canada, than has been previously recorded. In addition to the large range shift, host shift by MPB to jack pine (*P. banksiana* Lamb.) in Alberta has led to the entire Canadian boreal forest at risk of invasion by this beetle. Furthermore, MPB has recently spread into high-elevation pine habitats in North America due to ameliorating climatic conditions and has caused extensive mortality and declines of whitebark pine (*P. albicaulis* Engelm.), an ecologically important, keystone tree species in subalpine forest ecosystems where pines are already threatened by the invasive disease, white pine blister rust (caused by the pathogen, *Cronartium ribicola* J.C. Fisch.), fire suppression, and climate change [9–11].

Climate plays a critical and primary role in the biology and range delimitation of MPB given its modulating influence on the beetle's developmental processes and population growth. Consequently, what impacts rising temperatures and abnormal changes in weather patterns due to climate change will have on MPB outbreak behavior and spread in North America is a matter of serious concern to researchers, forest practitioners, and policy makers. The purpose of this paper therefore is to review the direct and indirect roles of climate-related factors on the development, outbreak behavior, and spread of MPB and briefly supplement this information with some existing evidence regarding other drivers of MPB spatial patterns for a more holistic picture of beetle population dynamics. In addition, we highlight the role of the recent MPB outbreaks in causing potentially undesirable ecosystem changes, which exemplify the indirect consequences of climate change on conifer forests via MPB outbreaks. We begin by providing a nonexhaustive overview of the distribution, host range, biology, ecology, epidemiology, and dispersal behavior of MPB to put things in context, before delving into the focal objective of this paper in more detail.

## Distribution, host range, and outbreak history

The MPB is currently distributed throughout western North America from northwest Mexico in the south, through the Pacific Northwest of the United States of America (USA), to the Canadian provinces of British Columbia and Alberta in the north [12]. Black Hills, South Dakota and western Nebraska, USA [13], and Cypress Hills, Saskatchewan, Canada, form the eastern frontier of MPB range. Elevational occurrences (reaching >3000 m above sea level) of MPB vary by latitude [7]. Typically, MPB populations are found at progressively lower elevations at cooler, poleward latitudes than hotter, southern latitudes.

Most species of pine native to North America are suitable hosts for MPB [7, 12, 14, 15]. Lodgepole pine and ponderosa pine (*P. ponderosa*) are the main hosts, whereas high-elevation Great Basin bristlecone pine *P. longaeva* Bailey [16, 17] and foxtail pine, *P. balfouriana* Balf. [17] are least preferred. Jack pine, *P. banksiana*, in Canada's boreal forest is a novel host [6]. Furthermore, MPB has the capability to reproduce on the nonnative Scots pine, *P. sylvestris* L. [14], and, in rare instances, native nonhosts such as interior hybrid spruce, *Picea glauca* (Moench) Voss × *engelmannii* Parry ex. Engelm. [18, 19], although there is no evidence to suggest that nonhosts are notable sources of population growth for MPB.

Tree mortality events due to MPB are visually apparent on the landscape and discernible as reddened trees one year after the actual tree death (Fig. 1C). The dead trees eventually turn gray after a few years (Fig. 1D). Aerial overview surveys using fixed-wing aircrafts or GPS-fitted helicopters are typically used to delineate the areal extents of MPB infestations. Available data suggest that two major peaks in MPB-caused tree mortality occurred in the USA in the past 40 years: first during the time period 1980–1981 (1.94 MHa/year) and second in the year 2009 (3.6 MHa) [20]. The latter peak was part of the outbreak that started in the early 2000s that covered at least ~122,000 ha per year between 2000 and 2019 [13]. During the same 40-year period in western Canada, large-scale MPB outbreaks were observed in the early 1980s and early 2000s in British Columbia [21]. More than 10 MHa of forested area was

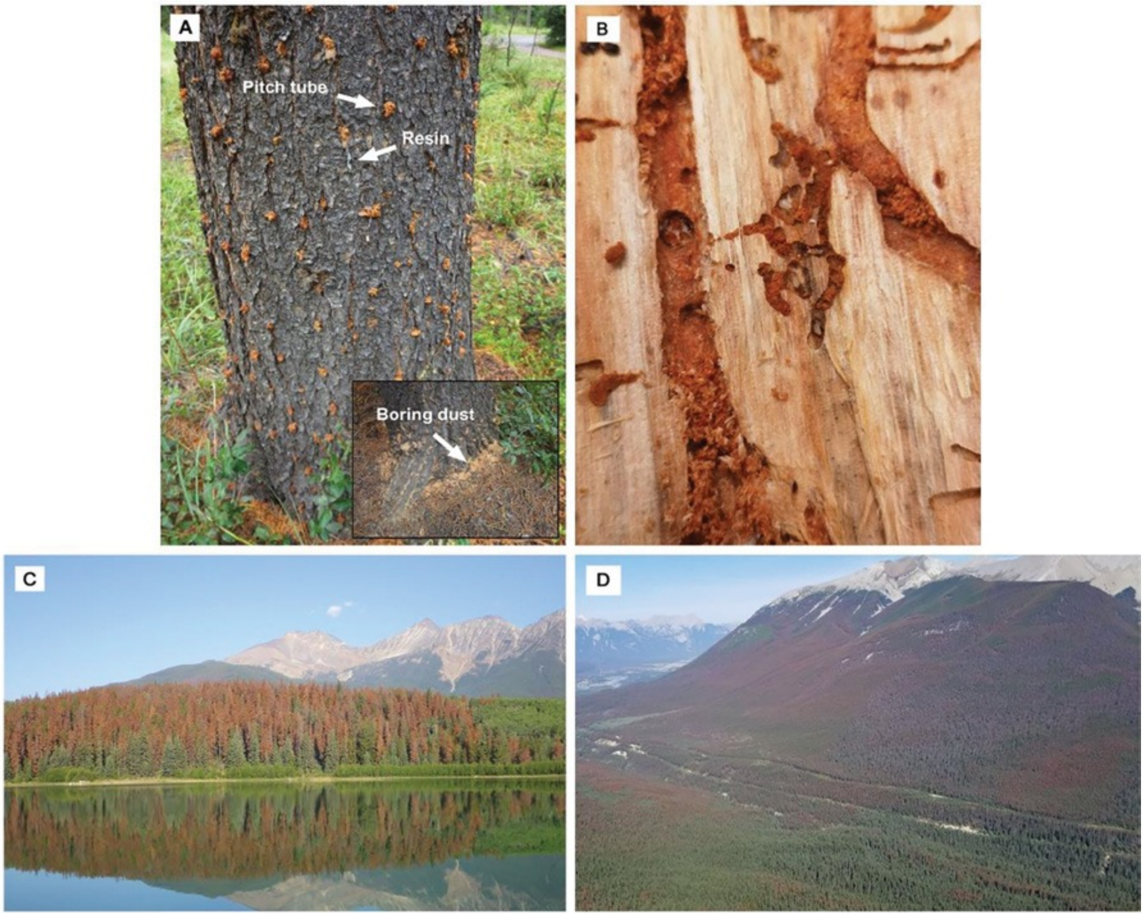


**Figure 1.** Signs of attacks by the mountain pine beetle (MPB) across different scales. A) Beetle attacks on a lodgepole pine tree. Note the dust on the tree and on the ground as beetles bore through the bark, and the tree exuding toxic resin in response to the beetle attack. B) Tunnels and larval gallery. C) MPB-killed trees (in red). D) MPB-affected landscape showing live (green) and dead (red, gray) pine trees. Beetle-killed trees appear red in about a year and then gray as the needles falloff. Images by Jim Weber and Brett Roger, Natural Resources Canada.

damaged in 2007 alone (Canada's National Forestry Database, http://nfdp.ccfm.org/en/index.php). Total area affected by MPB in British Columbia between 2000 and 2019 ranged from 113,781 ha to 10 Mha. In Alberta, more than 205,015 ha of forests were damaged by MPB each year between 2015 and 2018 (Canada's National Forestry Database).

## Life history, host selection, and ecology

Under warmer climates, MPB completes its life cycle in one year (univoltinism), normally starting with adults laying eggs in the phloem in the current year summer to progeny emergence as adults in midsummer of the following year. In areas that are too cold for optimal development of MPB, the life cycle could extend to 2 years (semivoltinism). Furthermore, combinations of different generation times, that is, univoltinism and semivoltinism and univoltinism and "fractional" voltinism (<1 year), have also been observed at some sites in the USA [22]. Host selection by MPB adults involves both visual cues and semiochemical-based attraction to host trees [23, 24] and active avoidance of nonhost chemical cues [25]. MPB initially attacks trees that are weakened or stressed (e.g., drought) or whose defenses have been compromised by other harmful agents such as secondary bark beetles or root diseases. Beetles colonize trees of all ages, although larger diameter trees are traditionally attacked often as they offer better fitness benefits, for example, higher brood production in trees with greater phloem thickness [26].

Upon landing, the females bore through the bark, start feeding, and construct vertical galleries in the phloem and outer sapwood [27] (Fig. 1A and B). This invasion disrupts the resin ducts and the attacked tree attempts to flush out the beetle in the resin flow. "Pitch tubes," that is, beetle entrance holes lined with tree resin and boring dust, are one of the characteristic signs of an MPB attack (Fig. 1A). The females release a potent aggregation pheromone, *trans*-verbenol, which in combination with host monoterpenes (e.g., α-pinene) attracts conspecifics of both sexes [28, 29]. Males release *exo*-brevicomin that further aids in tree colonization. Large numbers of beetles attacking the tree *en masse* result in tree death. The antiaggregation pheromone, verbenone, prevents overcrowding.

Mating and oviposition occur in the galleries. Most females lay up to 75 eggs (in some instances, more than 250 eggs) that hatch in less than a week between 18 °C and 27 °C [30, 31]. There are four larval instars in the MPB life cycle [27]. Larvae are typically exposed to extreme cold temperatures in the winter, but physiological adaptations (e.g., production of glycerol) allow population persistence [32]. Although all four larval instars may be observed during the winter [32], the third and fourth instar larvae constitute a higher proportion of overwintering larvae [32] and are less prone to cold-induced mortality than the early instars [33]. The fourth instar larvae pupate in the spring and the adults emerge from early June through early September depending on the location. In high-elevation regions or northern-most latitudes in the beetle's range where cooler climates persist during summer, slower rate of development results in a semivoltine life cycle.

MPB is associated with different species of ophiostomatoid fungi. These include *Grosmannia clavigera* (Robinson-Jeffrey and R.W. Davidson) Zipfel, Z.W.de Beer and M.J. Wingf., *Ophiostoma montium* (Rumbold) Arx, *Leptographium longiclavatum* S.W. Lee, J.J. Kim and C. Breuil, and *Ceratocystiopsis* spp. [34], although the composition of niche-sharing mutualistic fungal species differs spatially across the northern range of MPB [35]. The former two (i.e., *G. clavigera* and *O. montium*), however, are more consistently associated with MPB and are typically carried in maxillary exoskeletal invaginations known as mycangia and/or on the beetle exoskeleton [36–38]. As MPB colonizes the tree, multiple species of fungi may be introduced into the phloem [39]. The introduced fungi weaken the tree by seriously disrupting the transport of water and nutrients [40]. The main fungal associate, *G. clavigera*, aids in host colonization by enzymatically modifying or degrading host monoterpenes [41–43]. In addition, *G. clavigera* and *L. longiclavatum* may help MPB accrue nutritional benefits for its offspring as these fungi enhance nitrogen content at the site of infection in the phloem [44]. Hence, MPB-fungi association is mutually beneficial as it helps the beetles to overcome host tree defenses and live in a nutrient-rich subcortical environment, whereas the adult beetles help spread the fungi to new trees [45, 46].

## Population dynamics and dispersal

Spatial patterns of host colonization and the extent of damage caused by MPB depend on population-level behavioral traits that permit classification of populations into four distinct epidemiological phases: endemic, incipient-epidemic, epidemic, and postepidemic [12]. In the endemic phase, MPB population densities are not high enough to kill a healthy tree, so they are confined to weakened or senescent trees [12, 47, 48] and often cohabitate and compete with other subcortical feeding insects, including bark beetle species, within host trees [48]. Endemic populations transition to incipient-epidemic levels under favorable climatic conditions. In the incipient phase, beetle densities are adequate to kill a vigorous, large-diameter tree and the attacked tree clusters are limited to within a stand [12, 49]. Under conducive, multiyear temperature and precipitation regimes, scattered incipient-epidemic populations over a susceptible, highly connected host landscape become synchronous and eventually coalesce to form epidemic populations that infest wide swathes of pine forests [1, 12, 50]. The final postepidemic population phase is characterized by a waning population that may suffer high mortality from a variety of causes including host exhaustion; exposure to intense or untimely cold; natural enemies; and intraspecific competition [12].

Range shifts, invasions, and the expansion of MPB outbreaks at all spatial scales depend on dispersal of MPB adults to newer locations. Although MPB adults are not strong fliers, they are capable of relatively long flights, unassisted by wind, of up to 24 km at speeds of nearly 2 km/h [51]. The weak flying capability of MPB is inconsistent with the observation of beetle flights over distances far beyond their self-powered flight capabilities: in some instances, adult beetles have been observed at distances ranging from 80 km [52] to a few hundred km [53] from their source populations. This incongruence highlights that MPB exhibits at least two types of flight behaviors: the first, and the most studied, is dispersal below the forest canopy, which involves flights that are typically 50 m or less from the brood tree [54, 55]. The second is the above-canopy, passive long-distance dispersal.

Only a small proportion of MPB adults fly over the canopy [54]. Passive transport of the beetles above the canopy and into the atmospheric boundary layer occurs when turbulence underneath the canopy and updrafts created by low-pressure systems co-occur with MPB adult emergences [56]. Advective air currents then carry the beetles tens or even hundreds of kilometers from the source location [52, 57]. Analysis of the topographical features associated with the establishment of MPB populations dispersing long distances has revealed higher establishment in canyons and valleys than in regions with flatter terrain [58]. An assessment of the relative frequency of new population establishment by long- and short-distance dispersal in the recent outbreak in British Columbia suggests that short-distance dispersal was responsible for the establishment of most new MPB populations [59]. However, given that only a small proportion of adult beetles are transported above the canopy [54], the estimated proportion of new infestations (~15%) arising from long-distance dispersal events [60] in British Columbia from 1999 to 2007 [59] is quite significant. Long-distance dispersal likely has a disproportionate effect on the speed of MPB range expansions [61].

## Factors influencing MPB outbreak occurrences and landscape patterns of spread

Outbreak development and spread of MPB in forests dominated by its principal hosts (e.g., lodgepole pine) are governed by multiple factors, importantly climate and host susceptibility, that interact across scales to determine the eventual beetle distribution patterns on the landscape or regional level. Climate influences MPB population dynamics in three main ways: first, directly via its influence on the biological processes of MPB such as stage-specific survival, development rate, seasonality, and voltinism; second, by synchronizing beetle populations over the landscape and facilitating range expansions; and third, indirectly by influencing tree growth and susceptibility to MPB. Although climate will influence the tritrophic interactions, the outcomes of these complex processes are little understood. The following sections specifically address the following questions: 1) What is the direct and indirect role played by climate and weather in the biology, phenology, and outbreak behavior of MPB?; 2) What factors drive MPB population dynamics and spread?; and 3) What is the evidence that changes in climate have been implicated in the large-scale eruptions and spread of MPB?

## Direct influence of climate on MPB biology and landscape population dynamics

Temperature plays a critical role in the population success of MPB by regulating the development and timing of different life stages. Each life stage (egg, larval instars, pupa, and teneral adult) has a corresponding set of minimum (>5.6 °C – >15 °C) temperature requirements that are adaptively timed for the environmental conditions that the life stages will likely encounter [31, 62, 63]. Temperature-dependent growth behavior allows MPB, which lacks an obligate diapause, to complete its life cycle in about a year and ensures synchronicity in the emergence of adult beetles during a narrow temporal window in the summer [62]. For instance, lower temperature thresholds for development of eggs and earlier instars contribute to their faster growth in the fall. This allows temporal synchronicity with contemporaneous later instars such that a greater proportion of cold-tolerant larvae of similar ages overwinter [62, 64]. Fourth instar larvae have a higher temperature threshold for development than other life stages such that pupation only occurs in early summer when temperatures are high enough for larvae to pupate [65]. Synchronous adult emergence facilitates the possibility of a summer mass attack and rapid colonization that is needed to overwhelm the host defenses and kill the tree. In its historical range, MPB development rate is locally adaptive and latitude dependent such that, in northern climates where the temperatures are cooler and the growth period shorter, the beetles develop faster than the conspecifics at more southern latitudes [22, 66]. Cooler conditions delay MPB development in inhospitable areas such as high-elevation pine forests, and the consequent maladaptation of the different life stages to prevailing temperatures leads to beetle mortality and a decline in local populations [67].

Beetle flight occurs when summer temperatures are above ~16 °C [68] and declines at prolonged periods of high temperatures of 38 °C or above [68]. MPB flight is delayed and occurs over a shorter period of time in northern latitudes than more southern locations; however, physiological adjustments to local environments enable successful colonization [69]. Following host colonization, oviposition and gallery construction are influenced by temperature. The number of eggs laid per day, the number of eggs laid per unit gallery length, and daily gallery construction rate are positively related to increases in temperature [70].

Winter temperatures play a very important role in the survival and epidemiology of MPB. This species is intolerant to freezing and thus has developed adaptive mechanisms and strategies to avoid cold injury in the larval stage during the winter. Eggs and pupae are sensitive to cold and do not have overwintering capability [71, 72]. Temperatures below −23 °C kill the eggs [72]. Prolonged exposure to subzero temperatures above their supercooling points (SCPs), that is, temperature that triggers ice formation in insect body fluids (approximately −20 °C for eggs and − 18 °C to −20 °C for pupae), can cause significant mortality of these life stages [71, 72], although low-temperature effects on different MPB life stages can vary geographically [73]. In the case of the larvae, SCPs (< −35 °C; [32]) are much lower than the eggs, pupae, and teneral adults [71–73]. Whereas complete larval mortality is believed to occur below −40 °C, larvae exposed to subfreezing temperatures just above the SCP recover and survive [32]. As temperatures drop in the fall, physiological changes and cold-adaptive mechanisms are triggered in the larvae. They stop food consumption, remove agents that facilitate formation of ice crystals in the body (ice nucleators), and accumulate glycerol and other antifreeze compounds that enable them to counter the risk of freezing-induced death [64]. The proportion of cold-hardened or cold-susceptible larvae varies by season with a greater proportion of larvae possessing freeze protectants in the winter, while larvae in the fall or spring are cold susceptible [64]; hence, an unseasonal cold snap can cause serious mortality of MPB [74, 75]. Indeed, the average duration of a cold snap, the magnitude of temperature drop, and the frequency of extreme cold days negatively impact occurrences of MPB outbreaks [74].

Fluctuations of widely separated populations including insects that are under the influence of similar stochastic environmental forces and identical local density-dependent mechanisms are expected to be synchronous ("Moran effect") [76, 77]. In the case of MPB, although the local dynamics assumption may not strictly hold up given spatial heterogeneity and human-caused alterations to forests, weather does play a key role in synchronizing spatially disjoint beetle populations on the landscape [50, 77–79]. This weather-mediated rise and fall of populations is especially prominent during epidemic years in western Canada when population synchrony is observed over large distances (up to 900 km), while pre-epidemic, incipient populations are out of sync with each other over 200 km [50]. Although dispersal may facilitate MPB synchrony locally, it is not a major driver at regional scales [77]. Synchrony may also differ based on which pine species predominates on the landscape [79]. Actions of other agents such as parasitoids, predators, and competitors in modulating fluctuations of distant MPB populations are likely important, but their influences have not been adequately explored at a region-wide scale.

## Influence of climate on the host trees and niche-sharing fungi

Climate indirectly influences MPB population dynamics via its impact on tree physiology and growth. Persistent warm temperatures and prolonged dry spells are associated with extensive tree mortality and increased susceptibility of surviving host trees to bark beetle attacks [80, 81]. Severe drought-like conditions lead to breakdown of hydraulic conductance in the tree xylem due to embolism (i.e., air blockage) and loss of nonstructural carbohydrates resulting in tree death [82]. Precipitation (or lack thereof) and temperature profoundly influence tree growth responses [81, 83, 84] and production of chemical defenses [85, 86]. Growth of MPB's principal host, lodgepole pine, for instance, is impacted by monthly/seasonal weather fluctuations and site characteristics [87]. Average summer temperatures and higher summer precipitation in the previous growing season enhance growth in lodgepole pine. In the case of high-elevation limber pine, precipitation and May and July minimum temperatures are positively correlated with higher growth [81]. Decreases in tree growth occur under hot and dry conditions as carbohydrate reserves are rapidly depleted due to stomatal closure [82, 88]. Water stress significantly reduces net photosynthetic rate, lowers resin flow especially under bark beetle attack, and alters terpene profiles in ponderosa pine making it susceptible to bark beetle colonization [85]. Not surprisingly, arid conditions prior to local outbreaks provide a trigger for MPB population transition and expansion [79, 89–91]. These discrete populations then coalesce, self-amplify, and continue their spread to new locations until the host trees are exhausted or an adverse cold weather event such as a cold snap knocks down MPB populations [92].

Niche-sharing fungi are responsive to changes in temperature and soil moisture conditions [93–95]. Population genetic variations of *G. clavigera*, *O. montium*, and *L. longiclavatum* exhibit adaptive spatial trends and are molded by multiple factors including temperature and precipitation [93]. These fungi are temperature sensitive and show peak growth at ~25 °C, although *O. montium* is more tolerant to higher temperatures (30 °C) than the other two species [93]. In contrast, *G. clavigera* and *L. longiclavatum* survive extreme cold temperatures, whereas *O. montium* does not [94]. Tree responses to water stress under fungal colonization vary by tree species. For example, water deficit increases lesion length of *G. clavigera* in jack pine over time, while the contrary is true for lodgepole pine [95]. Water deficit alters the concentrations of individual monoterpenes but not the total monoterpene emission in *G. clavigera*–challenged lodgepole pine and jack pine [86]. Similarly, in lodgepole pine x jack pine hybrids, water stress causes changes in the composition of volatile emissions in *G. clavigera*–colonized trees, some of which may facilitate greater colonization by MPB [96].

## Modeling climate-MPB relationships and climate change impacts

An improved understanding of the MPB system over the past decades has allowed researchers to incorporate the knowledge gained for predictive purposes, for example, to describe future climatic suitability, or to link the patterns of infestations with possible drivers (Tables 1 and 2). Although multiple approaches have been used to study MPB-environment relationships, two approaches, and their combinations, are noteworthy: first, process-based (or mechanistic) mathematical models that simulate temperature-driven progression of biological process(es) such as phenology [97], cold tolerance [64], and both phenology and MPB population growth rate [98]; and second, b) traditional statistical models that associated MPB spatial infestation patterns with relevant predictors (Table 1). Recently, ecological niche models (or species distribution models) that correlate point species occurrences with environmental variables using machine learning algorithms have been used to characterize MPB distributions [99]. Influence of weather and climate on MPB population dynamics has been strongly demonstrated in multiple studies [50, 74, 79, 89, 90, 99–107] (Table 1). In general, warmer winters, average or above-average summer temperatures, and declines in precipitation strongly drive the spatiotemporal patterns of MPB infestations. However, it is important to note that other factors acting at small scales (e.g., stand and microsite) also play key roles in MPB epidemiology and short-range spread. Some examples of nonclimatic drivers of MPB infestations are presented in Table 2.

Climate change is occurring at a rapid pace and impacts of climate change on forests are manifesting, in part, as changes in disturbance regimes including insect outbreaks [108]. Given the temperature sensitivity of MPB, warming climates due to global change will undoubtedly render previously unsuitable locations suitable (or vice versa) for MPB establishment and population growth. Indeed, an assessment of past infestations of MPB in British Columbia, Canada, by Carroll et al. [104] showed that ameliorating climatic conditions helped MPB to spread to previously unsuitable habitats within the province, whereas previously suitable habitats became unsuitable in early-to-mid 1990s potentially due to excessive warming. MPB expanded its latitudinal and elevation ranges between 1959 and 1996 in

Table 1. A sample of studies that associated climate and mountain pine beetle outbreak patterns and/or modeled suitabilities under climate change.

| Region | Host species coverage | Time series | Mechanism(s)/drivers considered[1] | Warming-induced changes (past or projected future) | Analytical approach and reference(s) |
|---|---|---|---|---|---|
| USA and/or Canada | Range-wide | – | Adaptive seasonality | Geographically variable. Improved seasonality in northernmost latitudes/higher elevations (vice versa for southern latitudes and low elevations). | Process-based [97, 110, 111, 156] |
| North America and Europe | Range-wide | – | Phenology and winter mortality | High univoltine probability in more northern latitudes in the future. Canada's boreal forest less suitable in the long term. Receding ranges in the south. | Process-based [64, 113] |
| Canada | Range-wide | – | Temperature and precipitation: direct and derived variables | Northward and upward elevational shift. Improvement in climatic suitability at the southern edge of the boreal forest in the short term (2001–2030). | Derived index [104, 110, 157] |
| Colorado and southern Wyoming, USA | Lodgepole pine and ponderosa pine | 1996–2010 | Temperature and precipitation: direct variables; elevation; diameter at breast height; canopy cover; forest type; previous outbreak history | N/A | Graphical; spatial overlays [79] |
| Western USA | Lodgepole pine | Multiple (2010 and earlier) | Temperature and precipitation: direct and derived variables (annual, subannual, and seasonal) | N/A | Descriptive; graphical [89] |
| British Columbia, Canada | Lodgepole pine | 1959–1996 | Temperature, precipitation, aridity (Safranyik's climatic suitability model) | Northward and upward elevational shift. Expansion into previously unfavorable sites due to ameliorating climates over time. | Spatial overlays; linear/polynomial regression [104] |
| British Columbia, Canada | Lodgepole pine | 1972–1986 | Temperature: direct and derived variables; elevation; spatiotemporal infestations | N/A | Autologistic regression [100] |
| Western Canada | Lodgepole pine, lodgepole x jack pine; jack pine | 1990–1996 and 1999–2007 | Temperature: direct and derived variables; elevation; spatiotemporal infestations | Northward and upward elevational shift. | Autologistic regression [74] |
| Oregon and Washington, USA | Lodgepole pine | 1980–2006 | Temperature and precipitation: direct and derived variables (climate normal and weather; Logan's adaptive seasonality; Safranyik's climatic suitability index; cold tolerance); spatial coordinates; spatial and temporal infestations; soil moisture | N/A | Logistic regression [102] |

*Continued*

**Table 1.** Continued.

| Region | Host species coverage | Time series | Mechanism(s)/drivers considered[1] | Warming-induced changes (past or projected future) | Analytical approach and reference(s) |
|---|---|---|---|---|---|
| Greater Yellowstone Ecosystem (GYE), USA; North western USA | Whitebark pine | 1985/1986–2009 | Temperature and precipitation: direct and derived variables (cold-related variables; Logan's adaptive seasonality; precipitation; drought); host and habitat characteristics; spatial infestation patterns | Geographically variable. Climatic suitability projected to increase in the GYE by the end of this century. | Logistic regression [105,106] |
| Interior West USA (Colorado, Utah, Wyoming, Montana, Idaho) | Lodgepole pine, ponderosa pine, whitebark pine | 1960–1980 and 1997–2010 | Temperature and precipitation: direct and derived variables (annual, subannual, and seasonal); topographic variables | Shifts to higher elevations, but overall reduction in climatic suitability in the future. | Niche models [99] |

[1]We refer to "direct" variables as those that are calculated simply from meteorological data without further manipulations (e.g., mean temperature and annual precipitation). "Derived" variables are outputs from operations performed upon, or calculations made using, relevant meteorological data as inputs including specific calculations (e.g., cumulative degree days above a threshold, frost-free period start date, and drought indices), categorizations (e.g., binary), and model-based applications (e.g., adaptive seasonality).

**Table 2.** Some nonclimatic drivers of mountain pine beetle population dynamics and range movements.

| Drivers | Description and significance | References |
|---|---|---|
| Beetle population intensity | Spatial proximity to infested pine stands or previous occurrence(s) of infestations in the same area increases the risk of new infestations. | [74, 79, 100, 103, 106] |
| Forest attributes (% forest cover, diameter at breast height, basal area, stand density, stand age, host composition, forest type, pine volume) | Larger diameter trees with thick phloem on pine-dominated landscapes provide ideal breeding ground for MPB. | [11, 79, 103, 158–160] |
| Habitat fragmentation and connectivity | Less-fragmented, pine-rich landscapes may act as conduits for MPB spread especially in the epidemic phase, whereas transitioning populations may prefer fragmented landscapes. | [161–163] |
| Topography (elevation, aspect, slope, landscape characteristics) | Topography influences the microclimate, host tree distribution, exposure to sun light, soil moisture content, and creates barriers for movements, all of which influence MPB population dynamics. | [67, 74, 79, 100, 103, 164] |
| Fire history | Susceptibility of fire-affected lodgepole pine to MPB varies by fire severity and size. MPB prefers trees damaged by low-intensity fires. Large, high-severity fires may reduce tree susceptibility and impact beetle dispersal. | [161, 165–168] |
| Harvesting | Harvesting creates gaps on the landscape potentially reducing the spread and outbreak potential of MPB; however, edge effects may increase the risk of population growth by the transitioning populations. | [161, 163] |
| Forest legacy | Past beetle outbreak may drive changes in forest age-structure and species composition that may reduce future susceptibility to MPB. | [158, 169] |

British Columbia. In this case, above-average temperatures and dry weather patterns played an important role in expansive range shifts prior to the large epidemic that started in the late 1990s and early 2000s [90]. An overall improvement in climates in western Canada between years 1991 to 2010, coupled with an unprecedented eastward expansion by MPB in mid-2000s [58], has allowed populations to establish in jack pine forests of Alberta, Canada [5]. Climatic suitability will further improve in the short term in this region [109–111] and may facilitate eastward spread into the boreal forest, although several unknowns and uncertainties influencing population transition in the novel environment make assessments of spread rate difficult [5]. Warmer-than-normal temperatures and dry weather were correlated with MPB outbreaks in whitebark pine stands in ca. 1930 in Idaho, Montana, and Wyoming [7,112]. Recent increases in whitebark pine mortality in the USA have been attributed to increased climatic suitability (warmer winters and low summer precipitation), and future warming under climate change will likely lead to extensive whitebark pine mortality [105, 106].

Intermodel differences exist in the climatic suitability predictions for Canada when different aspects (climatic suitability, adaptive seasonality, and cold tolerance) are considered separately [110]. In totality, however, there is a low probability of MPB establishment and growth in central Canada in the future [110, 113]. Climatic suitability will shift northward with an overall decline in conditions needed for a univoltine life cycle over the long term in North America, although bivoltinism probability will increase in the southern USA and Mexico [113]. MPB populations do not currently exist in Europe. But if MPB were to invade Europe, conditions are presently suitable for both univoltine life cycle and cold survival in portions of central and northern Europe. However, by the end of the century, despite the probability of larval survival remaining high, climates favoring univoltinism will be restricted to the northernmost latitudes [113].

Range expansions by MPB into hitherto uninfested regions present a clear danger to pine ecosystems in North America under climate change (Fig. 2). Recent work sheds some light on different aspects of naïve pine-MPB interactions that should help us to understand and assess the risk of population growth and spread in novel pine habitats. For example, naive lodgepole pine trees are more susceptible to attack and are more reproductively suitable for MPB than lodgepole pines from regions experiencing outbreaks on a consistent basis as the former possess insufficient defenses to resist MPB attacks owing to a lack of coevolutionary history [114–116]. Many novel pine hosts such as jack pine, Scots pine, red pine (*P. resinosa* Sol. ex. Aiton), and eastern white pine (*P. strobus* L.) have higher relative levels of α-pinene, a precursor to the aggregation pheromone *trans*-verbenol that enhances attack rate, than lodgepole pine [15, 117]. Higher levels of α-pinene in jack pine could help MPB in host finding and colonization amid low pine density in the boreal forest [117, 118]. In addition, microbial symbionts and subcortical bacteria could further favor successful establishment in boreal jack pine [119–122]. In the case of whitebark pine, which is a qualitatively better host than lodgepole pine [123], defense mechanisms are

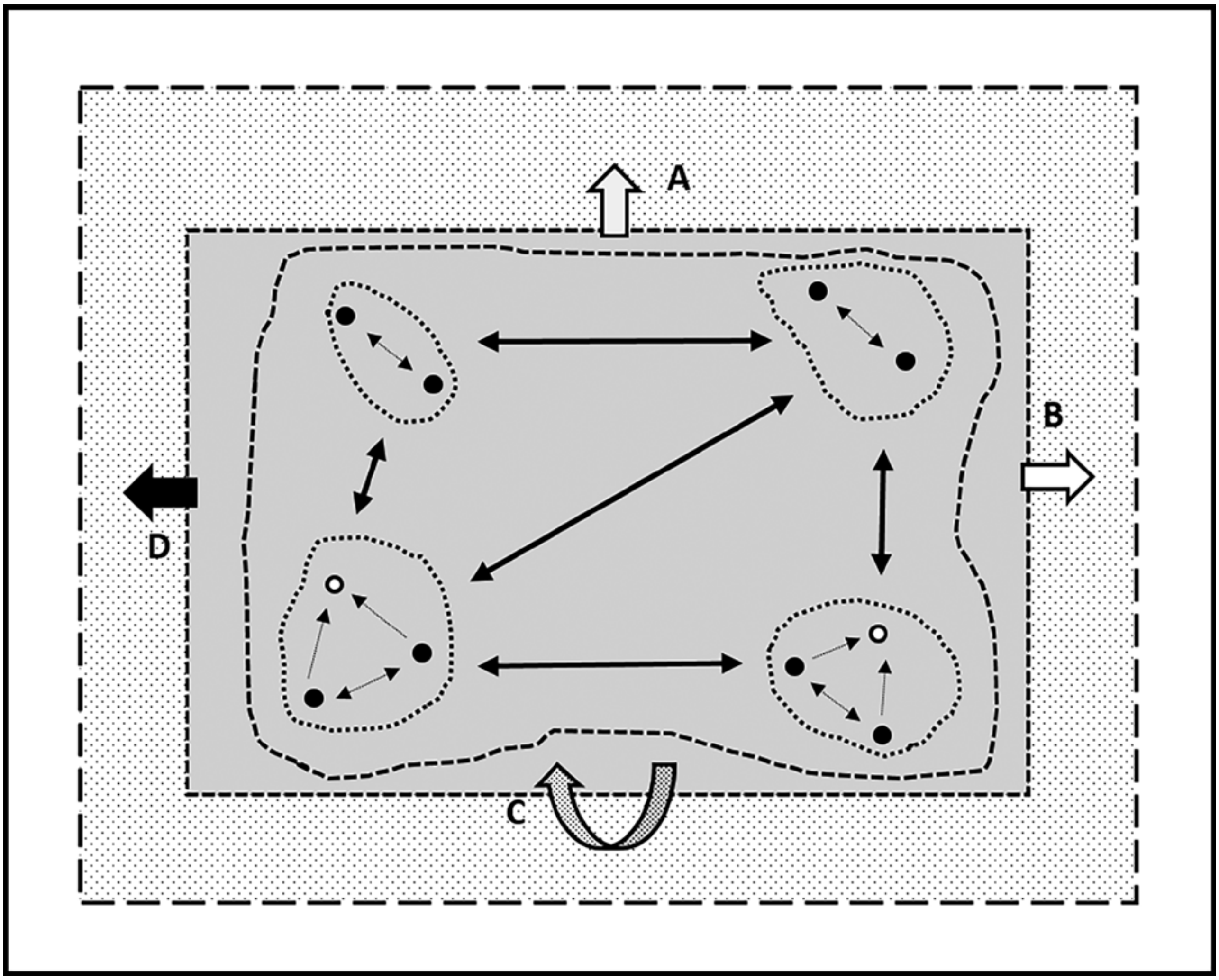


**Figure 2.** A simple conceptual diagram showing the potential fates of expanding mountain pine beetle (MPB) populations at the range edges leading into naïve habitats. The central shaded region represents the geographical area where the climate is historically suitable for MPB development. Here, incipient tree-killing populations (black dots), aided by favorable climatic conditions, spread to neighboring stands with endemic (white dots) or incipient populations [12]. These scattered populations eventually merge to form larger population clusters (dotted lines), that synchronize under favorable climates, to form a large outbreak (large dashed line) [12, 50]. The outer rectangular dotted region represents climate-induced extension in suitability zone over the geographic space (e.g., high elevations, northward latitudes) within the total extent of host pines in North America (outer solid bounding box) assuming some portion of the forest extent will remain unsuitable or become unsuitable under climate change. Expanding population fronts in newly suitable regions may survive as A and B) endemic populations (wide white arrows); C) cause occasional tree mortality and advance slowly, become endemic, or recede (due to competition, fire, disease impacts, intense cold event, proactive management, etc.); or D) transition to tree-killing populations that eventually erupt and spread rapidly due to increased reproductive capacity in naïve hosts [104, 114] (wide black arrow).

inadequate to counter beetle attacks and subsequent colonization by MPB and hence are at greater risk of being overwhelmed in the future [124].

There are several uncertainties that make the predictions of future MPB events in novel habitats under climate change challenging and therefore should be subjects of future investigations [5]. Adaptation strategies of MPB in novel environments, endemic-to-epidemic population transitions, direction and spread rates, behavior and phenology of niche-sharing associates and natural enemies, and among-tree variabilities in defense responses to beetle attack are topics that need to be studied more closely for a better assessment of invasion risk. Less is known about the influence of warming climates on complex gene-to-population-level changes in MPB and its interactions with other species in recently invaded or invasion-prone habitats. For instance, there are indications that subtle genetic changes in expanding beetle populations could potentially favor cold survival and contribute to further range expansion by MPB [125], but these interrelationships remain to be examined in more detail. Evolution of interspecific interactions in naïve pine habitats should be clarified. A potential mismatch in phenology among MPB and its main fungal symbionts, *G. clavigera* and *O. montium*, due to temperature-mediated predominance of a given fungal species under climate change, could affect the fitness and population success of MPB in new environments [126,127]. Range shifts of host pines in response to climate change add another layer of complexity to risk assessments. Geographic displacements of forest ecosystem niches and overall range contractions of key pine species are projected to occur by the end of this century [128–131]. The pace and direction of such range changes, however, will be determined by multiple factors such as trees' adaptive and migratory potentials, climate sensitivities of trees of different ages (e.g., saplings vs. older trees) especially at range edges, location\ecosystem characteristics, and alterations in disturbance regimes [129, 132–134]. In addition, faster adaptive responses of insects (given shorter life cycles), compared with the host trees, to subtle climatic changes and impacts due to other agents of mortality such as diseases and wildfires in pine ecosystems create further uncertainty in quantifying MPB distributions under changing climates. For example, whitebark pine and limber pine are already heavily affected by white pine blister rust [9, 135], and climate change may further aggravate disease-induced mortality of these pine species potentially affecting the future spread of MPB in high-elevation ecosystems.

## Ecological and economic impacts of MPB outbreaks

Tree mortality caused by MPB outbreaks impacts numerous biotic and abiotic components of the ecosystem. The preference of outbreaking MPB for larger diameter pines can result in the release of intermediate and suppressed trees below the canopy [136, 137]. Thus, in pine-dominated mixed stands, MPB outbreaks can precipitate a transition to dominant tree species that were formerly below the pine canopy [138]. In pure pine stands, suppressed and intermediate trees are released and exhibit rapid growth [136–138]. In pure lodgepole pine stands, the germination of pine seed is limited after MPB infestation in the absence of fire because thick carpets of bryophytes and other vegetation prevent seeds from reaching the mineral soil [139]. As a result, many lodgepole pine stands affected by MPB may not regenerate naturally to lodgepole pine [140] or their regeneration may be slow [141]. Nevertheless, increased forest diversity after MPB infestations may ultimately be beneficial in terms of system resilience as increased fire suppression and other land management practices homogenize landscapes and increase the risk of MPB outbreaks [142].

The impact of MPB outbreaks on subsequent fire severity is of societal concern and has been well studied. In Yellowstone National Park in the USA, a simulation study found no evidence that recent MPB infestation increased the likelihood of crown fires [143]. Rather, the probability of fires climbing to the crown was lower in recently infested stands with red or gray needles, than in green stands [143]. Analyses of fire severity data across a large component of the MPB's range in the Rocky Mountains did not find evidence that fire severity was impacted by the severity of MPB damage prior to fire in sites with red and gray attacked trees when fire weather was not extreme [144]. When fire weather was extreme, however, more severe MPB damage was associated with more severe fires in stands with red and gray attacked trees [144]. Quantification of the abundance and distribution of fuels after an MPB outbreak in Colorado, USA, in combination with fire simulations suggest that MPB infestation would reduce the likelihood of crown fires relative to stands that were not infested, but that surface fires would be more severe in MPB-infested stands [145]. Analysis of data on the severity of the Pole Creek fire in Oregon, however, found that the severity of crown and surface fires was negatively associated with the severity of MPB infestation that occurred 8 to 10 years before the fire [146]. Slightly different conclusions between studies suggest that the specifics of fuel accumulation, fuel moisture, and the time lag between MPB infestation and fire ignition are important factors in predicting the effect of recent MPB infestation on the severity of fires. However, a general conclusion shared across many studies is that prior MPB infestation tends to reduce the probability and severity of crown fires [143–146] in the decade after MPB outbreak.

In addition to fire, microbial decomposition of vegetation killed during MPB outbreaks is an important source of carbon after outbreaks collapse. When the carbon emissions due to decomposition, forest harvesting, and fire are simultaneously accounted, forests in British Columbia impacted by MPB are forecasted to transition from carbon sinks to carbon sources potentially aggravating climate change [8]. Moreover, forests in this province are predicted to continue acting as carbon sources in the decade following the collapse of the MPB outbreak [8]. The projected carbon emissions that resulted from the recent large-scale MPB outbreak in British Columbia accounted for 75% of the emissions due to forest fires from 1959 to 1999 in all of Canada [8].

The impacts of MPB infestation on wildlife depend on whether short-term behavioral responses or long-term demographic responses are considered. Ungulates may benefit from the increased forage due to proliferation of herbaceous understory species [147] that provide nutrition in the form of foliage or fruit after MPB outbreaks [148], an effect that will be manifested in ungulate demography. Conversely, in a short-term behavioral response, radio-collared elk avoided MPB-infested stands with many fallen snags due to movement difficulties [149]. Because caribou depend on arboreal lichen and avoid disturbed habitat, harvesting to mitigate the impacts of MPB may negatively impact caribou demography and cause them to avoid disturbed regions [150]. Human-caused landscape alterations such as road connectivity to MPB-killed stands for salvage harvesting may be playing a role in population decline (via increased access to hunters and natural predators) of moose in British Columbia [151, 152].

A management option currently implemented in numerous regions to minimize impacts on caribou and other wildlife is to create long-term harvest deferral zones. If large, such harvest deferrals will have important impacts on the progression of MPB outbreaks, forest dynamics, and fire risk—a fertile area for future research.

Many of the ecological impacts of MPB outbreaks can be translated to economic impacts when they directly impact commerce or industry. An effort to compute the long-term cost of the MPB outbreak in British Columbia suggests a loss of CAN$ 57.37 billion from 2009 to 2054 [2]. When impacts of MPB outbreaks on society are indirect, as is the case when ecosystems are impacted in ways that are difficult to assess from an industrial or resource extraction viewpoint, economic costs can be obtained by valuation of ecosystem services. Findings of ecosystem services analyses following the MPB outbreak in British Columbia suggest a trade-off between the direct economic benefit of salvage harvesting post-MPB in opposition to the economic value obtained from ecosystem services in the absence of extensive salvage harvesting [153].

At the time of writing, explicit accounting of the economic cost of MPB population management incurred by selected state, provincial, or federal governments did not appear in the scientific literature. Assessments of the impact of direct control by removal of individual infested trees in Cypress Hills Provincial Park in Saskatchewan, Canada, accounted for the aerial surveying cost, tree removal cost, and ground surveying costs [154]. To manage nearly 2.5 sq. km of forest in a single year of the ongoing MPB outbreak costed approximately CAN$ 200,000 [154]. In this case, higher costs may be attributed to spatially discrete, early-stage infestations in the study region that would entail greater survey and control efforts than would, for instance, areas with contiguous infestations. Alberta Agriculture and Forestry (AAF) coordinates similar ground surveys and removal of infested trees in high-priority regions of the province. Single MPB-infested trees and clusters of infested trees are felled and burned to destroy beetle progeny developing under the bark. In the 2015/2016 season, AAF coordinated tree removal and intensive ground surveys over 110.6 sq. km of forest (personal communication: Mike Undershultz, October 2020). Alberta-wide ground surveys and tree removal costed approximately CAN$ 17.4 million in 2015/2016 alone (personal communication: Erica Samis, October 2020). Overall costs were higher as this estimate does not include the cost of quality checking the work of the contractors who removed infested trees, the cost of coordinating level-two MPB control (logging and prescribed fire), and the cost of population monitoring.

## Summary

MPB is a bark beetle native to the pine forests of western North America. This eruptive herbivore exhibits pheromone-mediated mass aggregation behavior that overwhelms tree defenses, eventually killing the tree. Massive eruptions of this insect at the landscape and regional scale over susceptible pine-dominated habitats have been attributed to changes in climate. In this paper, we reviewed the direct and indirect influences of temperature and/or precipitation patterns on different aspects of MPB biology, ecology, and spatial outbreak patterns. We described the observed and expected changes in climatic suitability/range extensions under changing thermal regimes and concluded by examining how warming-driven MPB eruptions impact forests, ecologically and economically.

Accelerating pace of MPB-caused mortality, especially in borderline thermally suitable forests such as high-elevation pines, may be a "bioindicator" and "canary in the coal mine" of subtle environmental changes due to climate change [97, 155]. Indeed, even minor weather perturbations can upend sensitive MPB population dynamics and trigger a cascade of events that could culminate into a self-amplifying, potentially catastrophic, outbreak. Particularly worrisome are possible impacts of climate change in pine forests at the advancing fronts of MPB. Invasions in these "naïve" forests that have no or only intermittent coevolutionary history with MPB can have a destabilizing effect on ecosystem dynamics in the long term as they are at a greater risk of succumbing to outbreaks when supraoptimal climatic conditions arise. In the boreal forest, local adaptation and system equilibrium in the newly invaded habitat by MPB and its associates, combined with a warming climate, could set the stage for a rapid range expansion. When this could occur is difficult to predict and needs further investigation.

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